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IEOR 4578: Forecasting: A Real-World Application

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Technical Report:
A Chronologically Validated Framework for SKU-Level Demand Forecasting and Revenue Concentration Analysis

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1. Introduction

Accurate demand forecasting is a fundamental component of inventory management and retail operations. Weekly SKU-level forecasting supports replenishment planning, safety stock calibration, and the reduction of both stockouts and overstock risk.

This report develops and evaluates a weekly demand forecasting framework for a portfolio of operationally important retail SKUs. The analysis focuses on the top 250 revenue-generating SKUs with sufficient history, reflecting the subset of products most relevant for short- to medium-term planning. In addition to demand forecasting, the framework translates demand projections into projected revenue and portfolio concentration metrics to support managerial interpretation.

The objective is not simply to compare forecasting models in isolation, but to construct a forecasting pipeline that is both statistically defensible and operationally useful. Thus, models are selected using chronological validation and evaluated on a held-out test set divided into equal-sized periods.

The final aim is to identify a production-ready forecasting specification for each SKU and generate 12-week-ahead demand projections that can support tactical inventory and revenue planning.

2. Data Extraction

The dataset consists of historical retail transactions from a large European retailer. After cleaning, the data is used to construct weekly SKU-level demand and revenue series. Each transaction record contributes the following core fields:

- SKU identifier (StockCode)
- invoice date
- quantity sold
- unit price

The forecasting pipeline is built from transaction-level data but is ultimately modeled at the **weekly SKU level**, which is the operational decision frequency used throughout the report.

2.1 Workflow

Weekly SKU demand series were constructed through the following pipeline:

1. **Transaction cleaning**
   Cancellation invoices were removed, along with observations containing non-positive quantity or non-positive unit price. Records with missing invoice date, stock code, or invoice identifier were also excluded.

2. **Temporal indexing**
   Each transaction timestamp was converted to a Monday-anchored weekly period to ensure consistent time alignment across SKUs.

3. **Weekly aggregation**
   For each SKU, transaction quantities were aggregated into weekly demand totals. Revenue was constructed as quantity multiplied by observed transaction price and then aggregated to the weekly level.

4. **Temporal continuity enforcement**
   Each SKU time series was reindexed to a fixed weekly grid using a Monday frequency. Missing weeks within the observed span were filled with zero demand to preserve temporal continuity.

This preprocessing structure ensures that each SKU forms a consistent weekly sequence suitable for recursive multi-step forecasting and chronological evaluation.

3. Data Processing

3.1 Data Overview

The cleaned dataset contains multiple SKUs with heterogeneous demand profiles. These include:

- relatively stable demand SKUs,
- intermittent-demand SKUs,
- higher-variance SKUs.

As is typical in retail transaction data, the cross-section of SKU demand is highly heterogeneous and right-skewed, with substantial variation in both mean level and volatility.

The forecasting universe was restricted to the top 250 revenue-generating SKUs that also satisfied the minimum history requirement imposed by the train / validation / test design. This restriction improves operational relevance while ensuring that each included SKU has sufficient data to support stable estimation and evaluation.

3.2 Data Diagnostics

Exploratory diagnostics suggested three broad features of the panel:

- meaningful short-run autocorrelation in weekly demand,
- moderate non-stationarity in level for some SKUs,
- substantial heterogeneity in mean and variance across products.

To summarize cross-SKU demand volatility, the coefficient of variation,

$$CV = \frac{\sigma}{\mu},$$

was computed at the SKU level. This measure is used later in the appendix-style volatility segmentation to assess whether held-out performance differs across low-, mid-, and high-volatility products.

The available history spans roughly one annual cycle. This is sufficient for short-horizon forecasting, but it limits the ability to estimate recurring annual seasonality with high confidence.

3.3 Modeling Data Creation

Forecasting models were built from the weekly SKU demand series described above. For machine-learning models, supervised lag-based predictors were created directly from historical demand. In particular, the LightGBM specification used recent lagged weekly demand values as features:

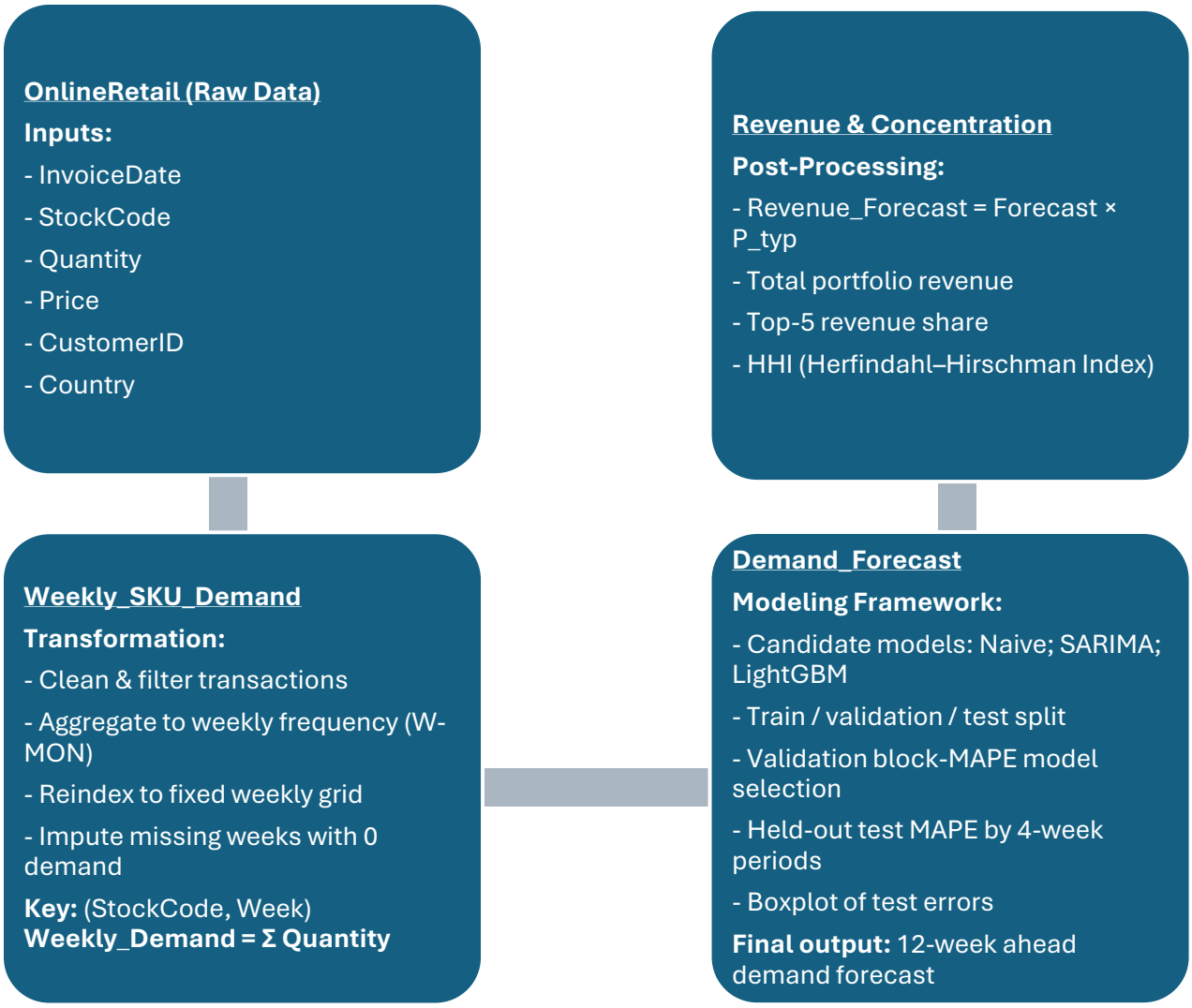
$$X_t = [y_{t-1}, y_{t-2}, \dots, y_{t-8}].$$

The target variable is weekly demand at time t. This lag formulation enables the model to learn nonlinear autoregressive patterns from recent demand history without introducing future information leakage.

Classical statistical benchmarks were also constructed from the same weekly demand series:

- **Naive**: last observation carried forward
- **SARIMA(1,1,1)**: non-seasonal autoregressive integrated moving-average model
- **LightGBM**: recursive lag-based gradient boosting model

All candidate models were therefore estimated from the same temporally aligned weekly demand foundation, ensuring comparability within the revised evaluation framework.



4. Target Variables

4.1 Demand per SKU

The main target variable is weekly unit demand for each SKU. Let

$$y_t^{(i)}$$

denote demand for SKU i in week t. This quantity represents the total number of units sold for SKU i, aggregated at the weekly level and indexed by calendar week.

Each SKU is modeled independently. This preserves SKU-specific demand structure and avoids imposing cross-product pooling assumptions in the baseline forecasting framework.

4.2 Weekly revenue per SKU

In addition to demand, weekly revenue is constructed as a secondary business output:

$$R_t^{(i)} = p_t^{(i)} \cdot y_t^{(i)},$$

where $p_t^{(i)}$ denotes unit price and $y_t^{(i)}$ denotes realized demand.

Because observed unit prices can vary across transactions and discount events, revenue forecasts are not produced by directly forecasting price. Instead, forecasted demand is translated into projected revenue using the median historical unit price for each SKU:

$$\hat{R}_{t+h}^{(i)} = \tilde{P}^{(i)} \cdot \hat{y}_{t+h}^{(i)},$$

where $\tilde{P}^{(i)}$ is median historical price and $\hat{y}_{t+h}^{(i)}$ is forecasted demand.

This approach improves robustness to temporary price noise and yields a transparent demand-conditioned revenue estimate over the 12-week projection horizon.

4.3 Revenue Concentration

Beyond SKU-level demand and revenue forecasts, the report evaluates portfolio concentration to assess whether projected revenue becomes increasingly dependent on a small subset of products.

Revenue share for SKU i at forecast horizon h is defined as:

$$S_{t+h}^{(i)} = \frac{\hat{R}_{t+h}^{(i)}}{\sum_j \hat{R}_{t+h}^{(j)}}.$$

Two concentration metrics are used:

4.3.1 *Top-5 Revenue Concentration*

To quantify managerial exposure to the leading products, the cumulative revenue share of the five largest projected SKUs at horizon h is computed as:

$$Top5_{t+h} = \sum_{i \in J_5} S_{t+h}^{(i)},$$

where J_5 denotes the set of five SKUs with highest projected revenue at that horizon.

4.3.2 *Herfindahl–Hirschman Index (HHI)*

To capture concentration across the full selected portfolio, the Herfindahl–Hirschman Index is computed as:

$$HHI_{t+h} = \sum_{i} \left(S_{t+h}^{(i)} \right)^2.$$

While Top-5 concentration provides an intuitive exposure metric, HHI summarizes how concentrated projected revenue is across the entire SKU set and is sensitive to shifts beyond only the largest products.

5. Predictive Variables

A predictive variable is any input used by the forecasting algorithm to estimate future demand. In this study, predictors are intentionally restricted to information derived from each SKU's own historical demand path.

5.1 Direct Variables

The primary direct input to the forecasting system is historical weekly demand. Although transaction-level price information is available in the raw data, price is not used as a direct regressor in the forecasting models. Instead, revenue is derived after demand forecasting using a historical price proxy.

No external regressors—such as promotions, holidays, markdown calendars, weather, or macroeconomic indicators—were included. Consequently, the predictive framework is purely autoregressive and extracts signal only from patterns embedded in past demand.

5.2 Derived Variables

Derived predictors were constructed to enhance predictive signal while preserving temporal causality.

5.2.1 *Lag Features (LightGBM)*

For the LightGBM specification, eight lagged demand variables were generated:

$$y_{t-1}, y_{t-2}, \dots, y_{t-8}.$$

These lag features allow the model to capture short-run nonlinear demand dynamics and interactions across recent periods.

5.2.2 *Differenced Demand (ARIMA)*

For the SARIMA model, first-order differencing was applied internally through the integrated component of the specification:

$$\Delta y_t = y_t - y_{t-1}.$$

This transformation helps stabilize the series prior to autoregressive and moving-average estimation and allows the model to represent short-run temporal dependence more effectively.

Because the final forecasting framework excludes exogenous predictors, all models operate within a purely autoregressive setting.

6. Pre Modeling

6.1 Data Structuring and Temporal Alignment

The dataset was aggregated to the weekly level for each SKU. To ensure temporal continuity and prevent irregular spacing in the time index, each SKU demand series was reindexed to a fixed Monday-anchored weekly frequency. Missing weeks were imputed with zero demand so that every SKU formed a continuous weekly sequence without gaps.

Formally, if $y_t^{(i)}$denotes weekly demand for SKU $i$in week t, the aligned series was constructed on a fixed grid using a Monday-based weekly index:

$$\left\{y_t^{(i)}\right\}_{t=1}^{T_i}, t \in \text{W-MON}.$$

Any missing calendar week within the observed span of a SKU was assigned:

$$y_t^{(i)} = 0.$$

This procedure ensures that recursive multi-step forecasting is performed on a temporally consistent sequence and prevents distortions caused by irregularly spaced observations. Each SKU was modeled independently to avoid cross-series contamination and to preserve SKU-specific demand structure.

6.2 Training, Validation, and Testing Split

Given the sequential nature of time-series data, random train-test splitting was not used. Instead, each eligible SKU series was partitioned chronologically into three disjoint segments: training, validation, and testing.

For each SKU, the final 28 weeks of observations were reserved for out-of-sample evaluation. These 28 weeks were divided into:

$$T_{\text{val}} = 12 \text{ weeks}, T_{\text{test}} = 16 \text{ weeks}.$$

Thus, if a SKU series contains T_i weekly observations, the split is:

$$\text{Train} = \{1, \dots, T_i - 28\}, \text{Validation} = \{T_i - 27, \dots, T_i - 16\}, \text{Test} = \{T_i - 15, \dots, T_i\}.$$

The training segment was used for initial model fitting, the validation segment for model selection, and the testing segment for final held-out performance reporting. This design preserves temporal ordering, prevents future information leakage into model fitting, and mirrors how the forecasting system would be deployed in practice.

To ensure a sufficient estimation window, only SKUs with adequate historical length were retained in the evaluation sample. Under this design, each included SKU had enough pre-validation history to support stable model estimation before the validation and testing windows began.

6.3 Validation and Held-Out Evaluation Framework

Model evaluation was conducted using a rolling-origin forecasting procedure within the validation and testing windows. At the start of each evaluation block, the model was fitted using only information available up to that point in time, and a multi-step forecast was generated for the next block. Once actual demand for that block became available, those realized observations were appended to the history before forecasting the following block.

For Deliverable 2, the 16-week test segment was divided into four equal-sized periods of four weeks each:

$$\text{Period 1, Period 2, Period 3, Period 4 (each of length 4 } weeks\text{).}$$

An analogous block structure was used on the 12-week validation segment, yielding three four-week validation blocks. Model choice for each SKU was based on validation performance, and final reported performance was computed only on the held-out test segment.

This framework satisfies three goals. First, it preserves chronological realism by ensuring that forecasts are always formed using past information only. Second, it aligns directly with the assignment requirement to report results over equal-sized test periods. Third, it provides a more stable basis for percentage-error evaluation than isolated weekly scoring, particularly for retail demand series containing low-volume observations.

Forecast outputs were constrained to be non-negative throughout to maintain operational realism.

6.4 Forecast Horizons

The primary operational deliverable in this project is a 12-week-ahead demand forecast for each SKU in the selected panel. This horizon was retained because it is long enough to support short- to medium-term inventory and replenishment planning while remaining within the range where historical autoregressive structure still contains useful predictive information.

At the same time, model evaluation for Deliverable 2 was not organized around the earlier horizon-by-horizon benchmarking design. Instead, accuracy was assessed over equal-sized four-week validation and testing blocks. This distinction is important:

- The **forecasting objective** is a 12-week demand projection for each SKU;
- The **evaluation design** is based on chronological validation and held-out testing using equal-sized 4-week periods.

This separation allows the report to remain operationally focused in the deliverables section while ensuring that model selection and performance reporting remain fully consistent with the assignment brief.

7. Modeling

7.1 Model Selection Rationale

To ensure robust benchmarking across heterogeneous SKU demand patterns, three conceptually distinct forecasting approaches were evaluated:

- Naïve (Last Observation Carry-Forward)
- SARIMA (ARIMA(1,1,1))
- LightGBM (Recursive Lag-Based Gradient Boosting)

These models were intentionally selected to represent three different modeling paradigms:

Model	Class	Captures	Bias–Variance Profile	Structural Assumptions
Naïve	Persistence baseline	Level continuity	High bias, low variance	Demand evolves slowly
SARIMA	Statistical time-series	Autocorrelation + trend	Moderate bias, moderate variance	Linear, stationary (after differencing)
LightGBM	Machine learning autoregression	Nonlinear lag interactions	Low bias, higher variance	Data-driven, minimal structural assumptions

This structured diversity allows performance comparisons that are not driven by model class alone, but by genuine predictive strength across heterogeneous SKU demand profiles under a common chronological validation and held-out testing design.

7.2 Model Properties and Theoretical Role

7.2.1 *Naïve Model (Last Observation Carry-Forward)*

The Naïve model provides the simplest possible forecasting benchmark, defined as:

$$\hat{y}_{t+h} = y_t$$

Under this specification, the most recent observed demand is carried forward across all forecast horizons. The model implicitly assumes demand persistence and does not estimate any parameters, impose structural form, or attempt to decompose trend or seasonality. As a result, it exhibits high bias but extremely low variance, making it resistant to overfitting and highly stable in short-horizon settings.

Its primary function within this study is to establish a baseline performance floor. Any model that fails to consistently outperform Naïve persistence offers limited incremental forecasting value. The Naïve model is particularly competitive for stable, low-volatility SKUs and very short forecast horizons, where structural change is minimal and demand continuity dominates.

7.2.2 *SARIMA (ARIMA(1,1,1))*

The SARIMA model represents the classical statistical time-series framework and is specified as:

$$(1 - \phi_1 L)(1 - L)y_t = (1 + \theta_1 L)\varepsilon_t$$

First differencing removes stochastic trend components, while autoregressive and moving-average terms explicitly model temporal dependence. Unlike the Naïve model, SARIMA assumes a structured linear relationship between current demand and past realizations.

This framework captures short-run momentum and mean reversion effects, providing a more flexible representation of serial correlation while maintaining interpretability. Relative to Naïve persistence, SARIMA reduces bias at the cost of moderately higher variance. It is particularly well suited to medium-volatility, structurally consistent demand series where linear autocorrelation dominates.

However, SARIMA assumes stable time-series dynamics and cannot model nonlinear lag interactions. Its performance may deteriorate under structural breaks or regime shifts. Within this benchmarking framework, SARIMA serves as **the classical statistical forecasting benchmark,** bridging simple persistence and modern machine learning approaches.

7.2.3 *LightGBM (Recursive Lag-Based Gradient Boosting)*

Supervised learning model using lag features:

$$y_t = f(y_{t-1}, y_{t-2}, \ldots, y_{t-k})$$

Unlike SARIMA, LightGBM imposes no parametric structure on the data-generating process. Instead, it uses a nonlinear tree-based ensemble to learn complex interactions between lagged demand values. This flexibility enables it to capture heterogeneous SKU behavior and nonlinear autoregressive patterns that linear time-series models cannot represent.

The model's functional form reduces bias by allowing complex temporal relationships, though at the cost of increased variance relative to SARIMA. Its predictive strength is particularly evident for high-volatility SKUs, where nonlinear dynamics and regime changes are more pronounced.

That said, LightGBM is sensitive to feature engineering quality and computationally more intensive. Within this study, it represents the **data-driven forecasting paradigm**, designed to exploit nonlinear temporal structure absent in traditional statistical models.

7.3 Validation Framework

All candidate models were evaluated using a chronological rolling-origin validation framework. For each SKU, the time series was split into training, validation, and testing segments, preserving temporal order and preventing future information leakage into model estimation.

The validation segment consisted of 12 weeks and was divided into three equal-sized 4-week blocks. The testing segment consisted of 16 weeks and was divided into four equal-sized 4-week periods. Within each block, forecasts were generated recursively using only information available up to that point in time. Once realized demand became available for that block, those observations were appended to the history before forecasting the next block.

This design serves two complementary purposes. First, it mirrors real deployment conditions by ensuring that forecasts are always formed using past information only. Second, it aligns directly with the assignment requirement to report performance over equal-sized testing periods rather than over a pooled rolling benchmark.

Model benchmarking was restricted to the top 250 revenue-generating SKUs, ensuring that evaluation remained focused on operationally material items while maintaining sufficient history for stable estimation.

7.4 Final Model Selection and Forecast Generation

Model performance on the validation segment was assessed at the 4-week block level. For each SKU, the model with the lowest validation block-level MAPE was selected as the final production specification. Once the preferred model had been identified, it was re-estimated using the full available history for that SKU and used to generate 12-week-ahead demand forecasts.

This SKU-specific model selection design reflects the heterogeneity of the demand panel and avoids imposing a single structural forecasting rule across the entire portfolio. Some SKUs exhibit smooth autoregressive structure, others show intermittent or more nonlinear short-run behavior. Allowing model choice to vary by SKU improves robustness and avoids imposing a single structural assumption across the full portfolio.

7.5 Evaluation Metric

Forecast errors are reported using Mean Absolute Percentage Error (MAPE) on a 0–100 scale. Because weekly SKU demand includes low-volume observations for which single-week percentage errors can become unstable, evaluation was conducted at the **4-week block level** rather than the isolated weekly level.

For SKU i and evaluation block b, realized and forecast demand were first aggregated within the 4-week block:

$$Y_{i,b} = \sum_{t \in b} y_{i,t}\,, \hat{Y}_{i,b} = \sum_{t \in b} \hat{y}_{i,t}.$$

Block-level absolute percentage error was then computed as:

$$APE_{i,b} = 100 \times \frac{\mid Y_{i,b} - \hat{Y}_{i,b} \mid}{\max\,(\mid Y_{i,b} \mid, 1)}.$$

The denominator floor of 1 unit ensures numerical stability when realized block demand is extremely small. Final block-level MAPE for a given validation or testing period is the average of $APE_{i,b}$ across SKUs in that period.

This metric preserves the required percentage-based interpretation while producing a more stable and operationally meaningful assessment of out-of-sample forecasting performance than isolated weekly SKU-level MAPE.

Forecast outputs were constrained to be non-negative to maintain operational realism.

7.6 Comparisons of Algorithmic Solutions

7.6.1 *Held-Out Test Performance by Equal-Sized Period*

Block Period	Test MAPE (%)	Median APE (%)	P90 APE (%)	P95 APE (%)
1	**91.66**	**35.64**	175.96	261.21
2	**61.73**	**43.76**	87.80	100.00
3	**71.98**	**38.46**	151.77	227.45
4	**59.56**	**35.10**	132.56	208.61

Table 7.6.1 summarizes block-level held-out test performance across the four equal-sized testing periods. Performance is weakest in Period 1, where test MAPE is 91.66, and strongest in Period 4, where test MAPE declines to 59.56. Periods 2 and 3 record intermediate values of 61.73 and 71.98, respectively. Because each period contains 250 SKU-block observations, the equal-weight average held-out block-level MAPE across the full test window is approximately 71.23.

Median APE is materially more stable across periods, ranging from **35.10** to **43.76**. This indicates that the higher mean percentage errors are driven by a right tail of difficult blocks rather than by broad forecast failure across the entire SKU panel. The tail remains most pronounced in Periods 1 and 3, where P90 and P95 values are higher than in Period 2 and, to a lesser extent, Period 4.

7.6.2 *Error Distribution Across Held-Out Periods*

The boxplot of block-level APE across the four test periods provides a complementary view of dispersion. Median errors remain relatively stable across all periods, with somewhat tighter distributions in Periods 2 and 4 and a wider upper tail in Period 1. This pattern is consistent with the aggregate MAPE table: the dominant difference across periods lies less in the center of the distribution than in the severity of the upper tail.

Operationally, this suggests that the forecasting system remains broadly stable across the held-out horizon, but certain four-week windows contain more adverse SKU-level realizations that inflate mean percentage error. Period 1 appears to be the most challenging test window, while Period 4 is the most favorable.

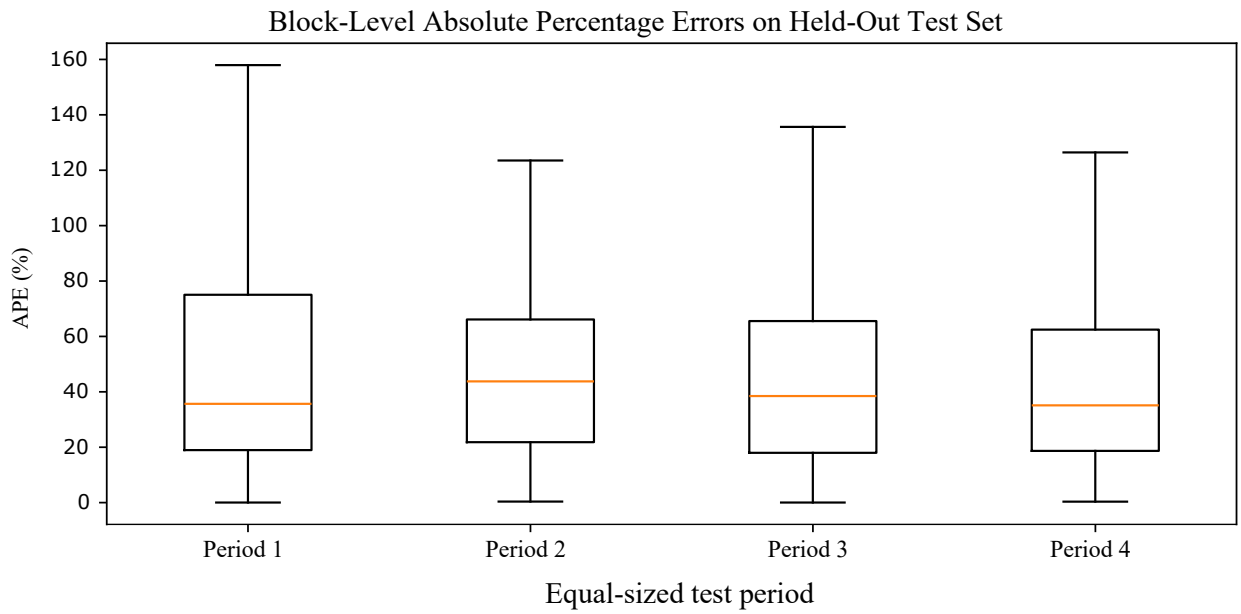


Figure 7.6.2: Block-Level Absolute Percentage Errors on Held-Out Test Set

7.6.3 *Model Selection Outcomes*

Figure 7.6.3 summarizes the final model selected per SKU under the validation block-MAPE criterion. SARIMA is selected for 98 of 250 SKUs (39.2%), LightGBM for 84 SKUs (33.6%), and Naive for 68 SKUs (27.2%). This confirms that SARIMA remains the dominant specification in the panel even under the revised Deliverable 2 evaluation framework.

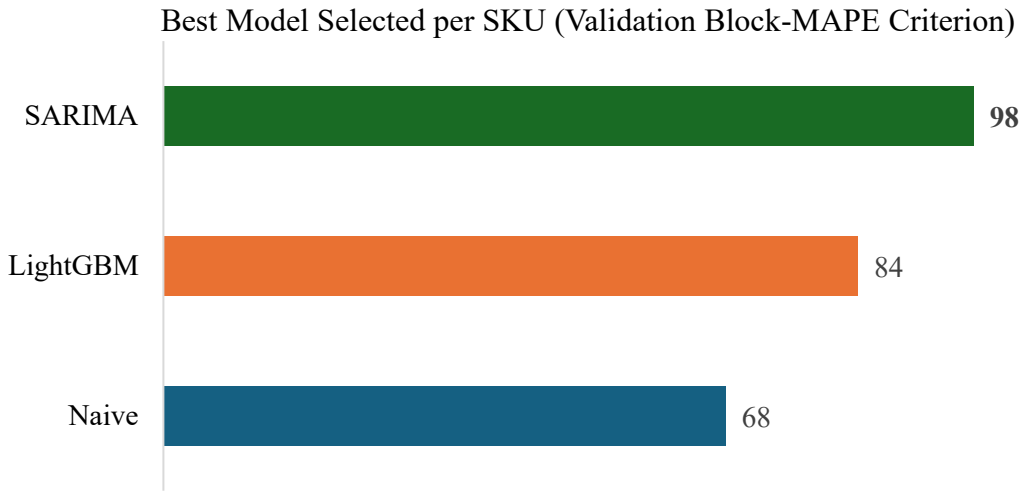


Figure 7.6.3: Final model selected per SKU (validation block-MAPE criterion)

The selection frequencies are operationally informative. They indicate that linear autoregressive structure is sufficiently persistent to dominate the largest share of the SKU universe, while LightGBM captures a substantial minority of products whose short-run dynamics appear less well represented by a purely linear specification. The Naive benchmark remains useful for a narrower subset of stable series but is selected least often overall.

7.6.4 *Held-Out Test Performance by Chosen Model*

A breakdown of held-out performance by chosen model further clarifies the structure of the final solution. SKUs assigned to SARIMA exhibit the most stable out-of-sample performance, with test MAPE remaining in a relatively narrow band across periods (**60.43**, **52.18**, **52.22**, **53.12**). LightGBM-selected SKUs also perform credibly, though somewhat less consistently, with period MAPEs ranging from **52.81** to **63.93**. In contrast, SKUs assigned to the Naive benchmark exhibit the highest period-to-period variability and the weakest performance in adverse test windows, including a particularly weak result in Period 1 (**184.67**) and a second large deterioration in Period 3 (**116.76**).

These comparisons are descriptive rather than experimental, since each chosen-model group corresponds to a different subset of SKUs rather than a common evaluation panel. However, they reinforce the broader conclusion that SARIMA provides the most robust and stable solution on the largest share of the panel, while Naive persistence is only suitable for a narrower subset of smoother demand series.

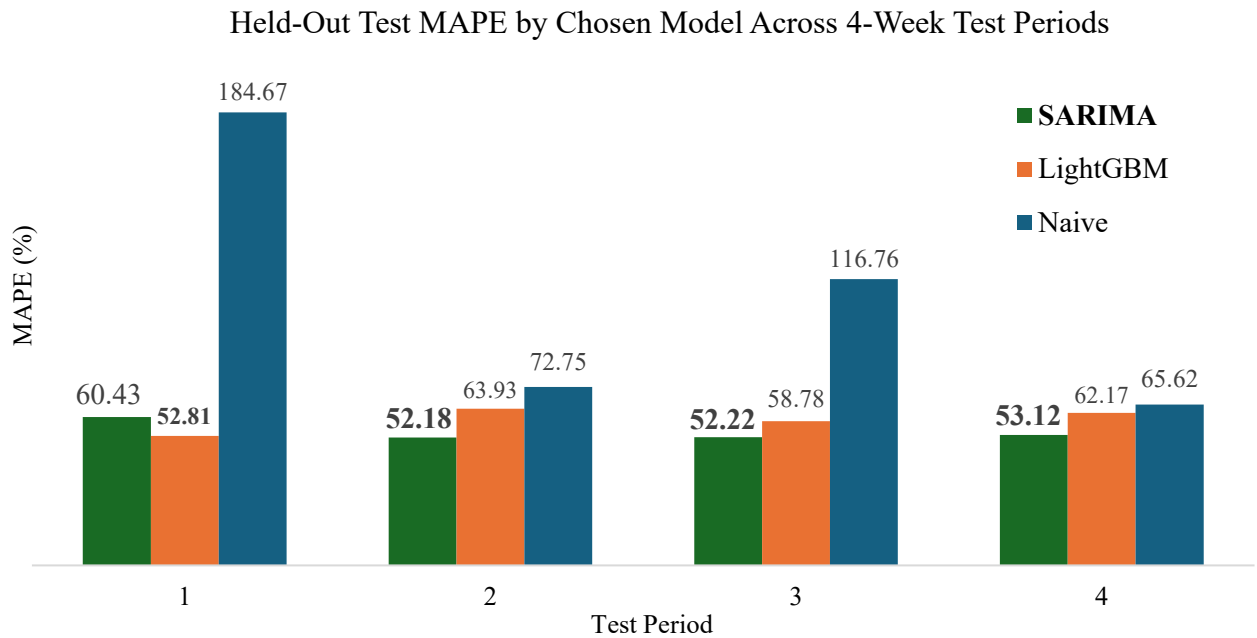


Figure 7.6.4: Held-out block-level test MAPE by chosen model across the four 4-week test periods.

7.6.5 *Interpretation*

Across validation performance, held-out test MAPE, and final model-selection frequencies, the evidence points to a consistent conclusion: SARIMA remains the dominant forecasting specification for the SKU portfolio. Its advantage is now expressed through lower validation block-MAPE, greater model-selection frequency, and the most stable held-out block-level performance.

LightGBM remains relevant and captures a meaningful subset of SKUs, indicating that nonlinear lag interactions add value for some products. However, its higher validation MAPE and less stable held-out profile suggest that nonlinear flexibility does not translate into a systematic portfolio-wide advantage. The Naive model remains operationally useful as a benchmark and for a limited

set of smoother series, but is selected least often and exhibits the greatest vulnerability to adverse testing periods.

Overall, the revised evidence suggests that structured linear autocorrelation remains the primary source of predictive signal in this retail SKU panel, while more flexible models provide selective rather than universal gains.

8. Deliverables

The primary operational deliverable is a 12-week-ahead demand forecast for each SKU in the selected retail panel. Forecasts are generated using the SKU-specific model selected under the validation block-MAPE criterion and re-estimated on the full available history prior to final projection.

The 12-week horizon balances operational relevance with statistical reliability. It is long enough to support short- to medium-term replenishment and inventory planning while remaining within the range where out-of-sample performance remains interpretable under the held-out evaluation framework.

SKU-level demand forecasts are translated into projected revenue using median historical SKU prices to reduce distortion from temporary pricing outliers.

The main forecast outputs include:

- **Final_Demand_Forecasts.xlsx** — SKU-level 12-week demand projections
- **Portfolio revenue summary table** — projected revenue and concentration metrics by horizon
- **Top-SKU composition figures** — projected demand and projected revenue concentration among leading SKUs

These deliverables provide an operationally interpretable forecast layer built on a statistically validated model-selection framework.

8.1 Forecast Demands

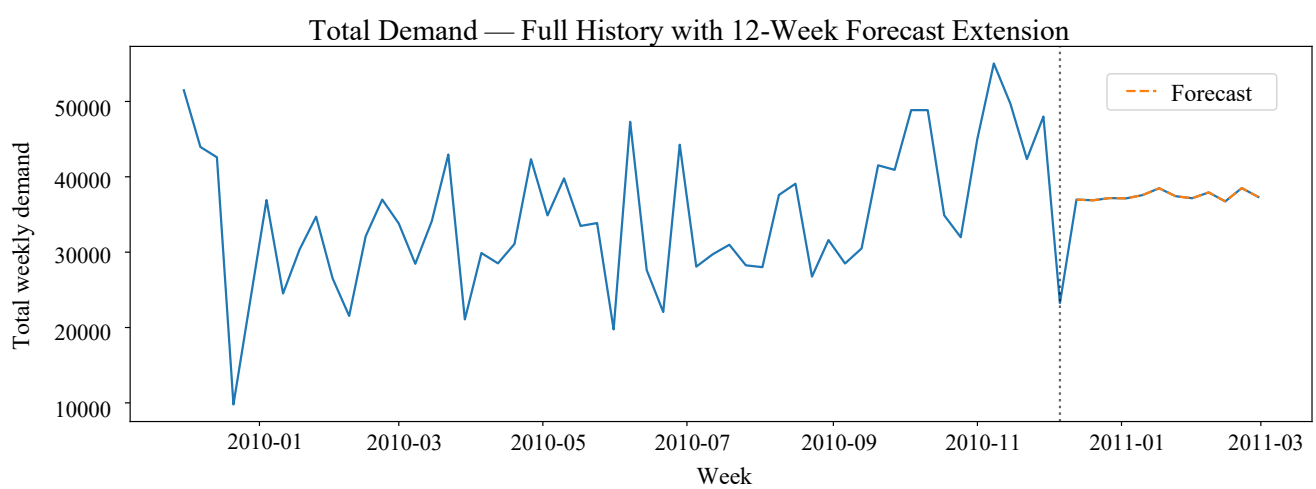


Figure 8.1: Aggregate weekly demand with 12-week forecast extension.

Figure 8.1 presents aggregate weekly demand across the selected SKU panel, extended by the final 12-week forecast. The forecast profile remains relatively stable across the projection window, with no evidence of explosive growth or sharp structural break. Relative to the historical series, the projected demand path appears smoother and remains centered within the recent range of observed aggregate demand.

This pattern is consistent with the broader modeling evidence from Section 7. The final selected models, particularly SARIMA on the largest share of SKUs, capture short-run autoregressive structure without generating unstable multi-step extrapolations. Overall, the forecast suggests continuation of prevailing portfolio-level demand conditions rather than a regime shift.

8.2 Portfolio Revenue Outlook

Summary of portfolio revenue characteristics by horizon

Horizons	Total Revenue	Revenue Concentration: Share of Top-5 SKU's	HHI
1	$67,931	18.3%	0.0135
2	$68,413	17.2%	0.0127
3	$68,533	15.9%	0.0113
4	$67,603	16.9%	0.0121
5	$70,708	17.8%	0.0133
6	$69,531	18.0%	0.0131
7	$70,481	16.6%	0.0127
8	$67,819	18.4%	0.0137
9	$69,105	17.9%	0.0132
10	$67,945	16.9%	0.0122
11	$70,933	17.4%	0.0125
12	$68,318	17.6%	0.0129

Table 8.2: Summary of portfolio revenue characteristics by horizon

Table 8.2 summarizes projected portfolio revenue and concentration metrics across the 12-week forecast horizon. Total projected weekly revenue remains broadly stable, fluctuating within a narrow band of approximately $67.6k to $70.9k. No pronounced upward or downward trend emerges over the forecast window, suggesting that the demand forecast translates into a relatively steady revenue outlook.

Revenue concentration remains moderate across horizons. The top five SKUs account for roughly 15.9% to 18.4% of projected weekly revenue, while HHI ranges from approximately 0.0113 to 0.0137. These values indicate that the projected revenue base remains diversified despite moderate dependence on the upper tier of SKUs.

Taken together, the results suggest that forecast variability affects overall revenue magnitude more than portfolio composition. Relative revenue composition remains fairly stable across the 12-week

horizon, and there is no evidence that concentration risk intensifies materially as the forecast extends.

8.3 Managerial Interpretation

The forecast outputs indicate a stable short- to medium-term demand environment, with no evidence of broad structural disruption at the portfolio level. Revenue remains diversified across SKUs, limiting exposure to single-product concentration risk, although dependence on the strongest-performing items remains economically meaningful.

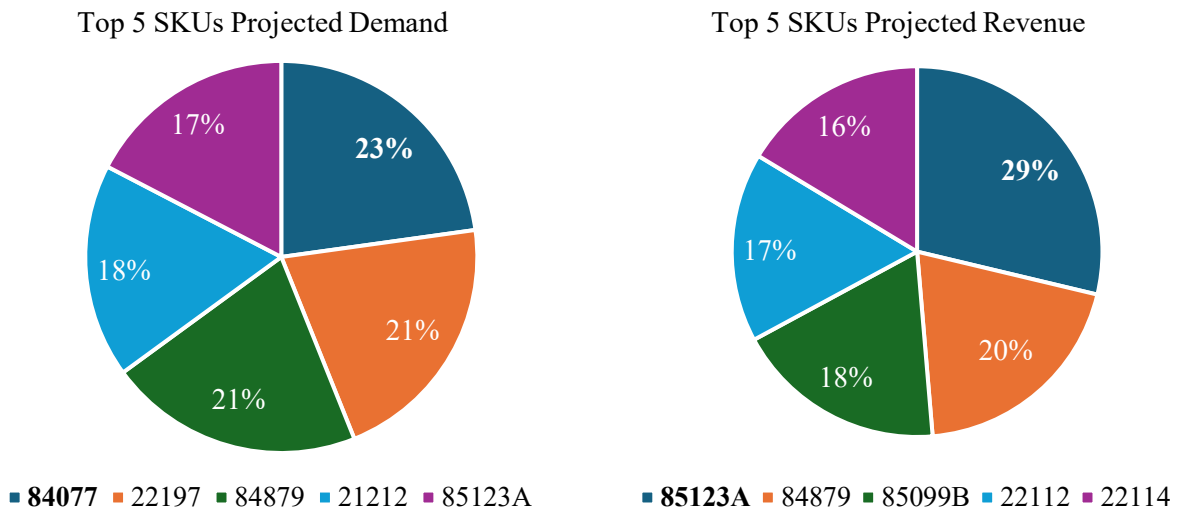


Figure 8.3: Top 5 SKUs by projected demand and projected revenue.

Figure 8.3 shows that concentration is more pronounced in projected revenue than in projected demand. In the top-five demand group, the leading SKU contributes approximately 23% of projected demand, whereas in the top-five revenue group, the leading SKU contributes about 29% of projected revenue. This suggests modest margin heterogeneity across products, with some SKUs contributing more revenue than volume alone would imply.

Operationally, this pattern indicates that inventory and replenishment risk is not concentrated across the full portfolio, but it is layered within the upper-performing SKU tier. The forecast therefore supports steady planning at the aggregate level, while highlighting the importance of monitoring top-SKU execution and revenue sensitivity more closely than the broader panel.

8.4 Agentic Forecast Retrieval Layer

To improve managerial accessibility, the forecasting pipeline was extended with a lightweight prompt-based forecast retrieval interface. Rather than manually inspecting forecast tables, a business user can submit a natural-language request containing a valid SKU identifier (StockCode), such as *"Give me the latest forecast for SKU 85123A."* The system then parses the prompt, identifies the requested SKU, maps it to the SKU-specific selected model, and retrieves the latest precomputed 12-week forecast outputs.

The retrieval layer is built directly on top of the final modelling artifacts produced by the pipeline, including the SKU-level model selection table, the horizon-level demand forecast table, and the revenue translation outputs. For a valid request, the interface returns a manager-facing summary

containing the selected model, validation block-MAPE, projected 12-week demand, projected 12-week revenue, and the week-by-week forecast path over the 12-week horizon. Figure 8.4 illustrates a representative example of this prompt-based interaction.

```
40]: example_prompt_1 = "What is the latest 12-week forecast for product 22197?"
     example_prompt_2 = "What is the 12-week demand and revenue outlook for product 84077?"

     print(get_latest_forecast_for_sku(example_prompt_1, agent_summary, agent_horizon))
     print()
     print(get_latest_forecast_for_sku(example_prompt_2, agent_summary, agent_horizon))
```

```
Forecast retrieval result for SKU 22197
--------------------------------------------------
Selected model: SARIMA
Validation block-MAPE: 27.63
12-week projected demand: 16977.8
12-week projected revenue: $14,431.16
Median historical price proxy: $0.85

Weekly forecast breakdown:
Week  1: Demand = 1499.6, Revenue = $1,274.63
Week  2: Demand = 1392.9, Revenue = $1,183.97
Week  3: Demand = 1410.9, Revenue = $1,199.28
Week  4: Demand = 1407.9, Revenue = $1,196.69
Week  5: Demand = 1408.4, Revenue = $1,197.13
Week  6: Demand = 1408.3, Revenue = $1,197.06
Week  7: Demand = 1408.3, Revenue = $1,197.07
Week  8: Demand = 1408.3, Revenue = $1,197.07
Week  9: Demand = 1408.3, Revenue = $1,197.07
Week 10: Demand = 1408.3, Revenue = $1,197.07
Week 11: Demand = 1408.3, Revenue = $1,197.07
Week 12: Demand = 1408.3, Revenue = $1,197.07

Forecast retrieval result for SKU 84077
--------------------------------------------------
Selected model: SARIMA
Validation block-MAPE: 20.55
12-week projected demand: 18325.6
12-week projected revenue: $3,848.38
Median historical price proxy: $0.21

Weekly forecast breakdown:
Week  1: Demand = 1653.3, Revenue = $347.19
Week  2: Demand = 1501.0, Revenue = $315.22
Week  3: Demand = 1518.8, Revenue = $318.95
Week  4: Demand = 1516.7, Revenue = $318.52
Week  5: Demand = 1517.0, Revenue = $318.57
Week  6: Demand = 1517.0, Revenue = $318.56
Week  7: Demand = 1517.0, Revenue = $318.56
Week  8: Demand = 1517.0, Revenue = $318.56
Week  9: Demand = 1517.0, Revenue = $318.56
Week 10: Demand = 1517.0, Revenue = $318.56
Week 11: Demand = 1517.0, Revenue = $318.56
Week 12: Demand = 1517.0, Revenue = $318.56
```

Figure 8.4: Example output from the prompt-based forecast retrieval interface. A natural-language request containing a valid SKU is parsed and mapped to the SKU-specific selected model and latest 12-week forecast outputs.

Operationally, this interface functions as a lightweight agentic layer: it automates prompt interpretation, routing, and forecast retrieval without requiring the user to manually search across output files. While the current implementation uses deterministic parsing rules rather than a full external LLM service, it satisfies the practical objective of enabling prompt-based access to the latest forecasts in a transparent and reproducible manner. A natural extension would be to replace the rule-based retrieval layer with a local LLM (e.g., Ollama) to support more flexible prompt understanding and product-description matching.

9. Model Limitations

While the forecasting framework demonstrates reasonable stability under chronological validation and held-out testing, several limitations should be acknowledged.

Limited Historical Horizon

The dataset spans approximately one annual cycle, which limits the ability to estimate recurring seasonal structure with high statistical confidence. This is particularly important for retail demand, where holiday effects, promotional cycles, and calendar-driven purchasing behavior may repeat annually but cannot be robustly identified from a single cycle. Longer historical coverage would improve seasonal estimation and reduce parameter uncertainty.

Absence of Exogenous Drivers

The modeling framework relies exclusively on autoregressive demand information. No external regressors—such as promotions, holidays, markdowns, weather, or macroeconomic indicators—were incorporated into the forecasting system. As a result, forecasts are well suited to capturing internal demand persistence but may respond weakly to structural changes or event-driven demand shocks that are not embedded in recent sales history.

Recursive Multi-Step Forecasting

The 12-week forecasts are generated recursively, so forecast error can propagate across the projection horizon as each step depends on prior predicted values. This is a standard feature of multi-step time-series forecasting, but it implies that longer-horizon projections are inherently less precise than short-horizon projections. The held-out test results partly reflect this challenge, particularly in periods where upper-tail block-level percentage errors widen.

Percentage-Error Sensitivity

Deliverable 2 required results to be reported using MAPE on a 0–100 scale. Although block-level aggregation substantially improved stability relative to single-week SKU-level percentage errors, percentage-based metrics remain sensitive when realized demand is very small. Accordingly, the reported block-level MAPE should be interpreted as a stable and assignment-consistent summary statistic, but not as a complete characterization of all aspects of forecast quality.

Restricted Evaluation Universe

Model evaluation was restricted to the top 250 revenue-generating SKUs with sufficient history. This improves operational relevance and estimation stability, but it means the reported performance does not necessarily generalize to the full SKU universe, particularly to very low-

volume or highly intermittent items. Extending the framework to the long tail of products would likely require different evaluation choices and potentially different modeling tools.

Revenue Translation Assumptions

Projected revenue was computed using the median historical SKU-level unit price. This improves robustness to transaction-level price noise and temporary discounts, but it assumes that short-run price structure remains broadly stable over the forecast horizon. The revenue outputs should therefore be interpreted as demand-conditioned revenue approximations rather than as fully dynamic price-demand forecasts.

Future work could incorporate longer histories, exogenous regressors, and probabilistic scenario analysis to improve responsiveness to structural shocks and to deepen medium-term planning value.

10. **Conclusion**

The revised evidence points to a clear conclusion: SARIMA remains the strongest overall forecasting specification for this SKU portfolio under the MAPE framework. It achieved the lowest validation block-MAPE, was selected most frequently across SKUs, and delivered the most stable held-out performance across the four equal-sized test periods.

At the portfolio level, the final 12-week forecasts indicate a broadly stable short- to medium-term demand and revenue outlook. Concentration metrics remain moderate across horizons, suggesting that forecast risk is more closely tied to top-SKU execution than to broad structural instability across the full panel.

Overall, the framework provides a statistically defensible and operationally useful basis for short- to medium-term planning. The most natural next improvement would be to incorporate exogenous demand drivers and longer historical coverage to better capture structural shifts and seasonal effects.

11. Appendix

Appendix A1: Revenue Outlook by Horizon

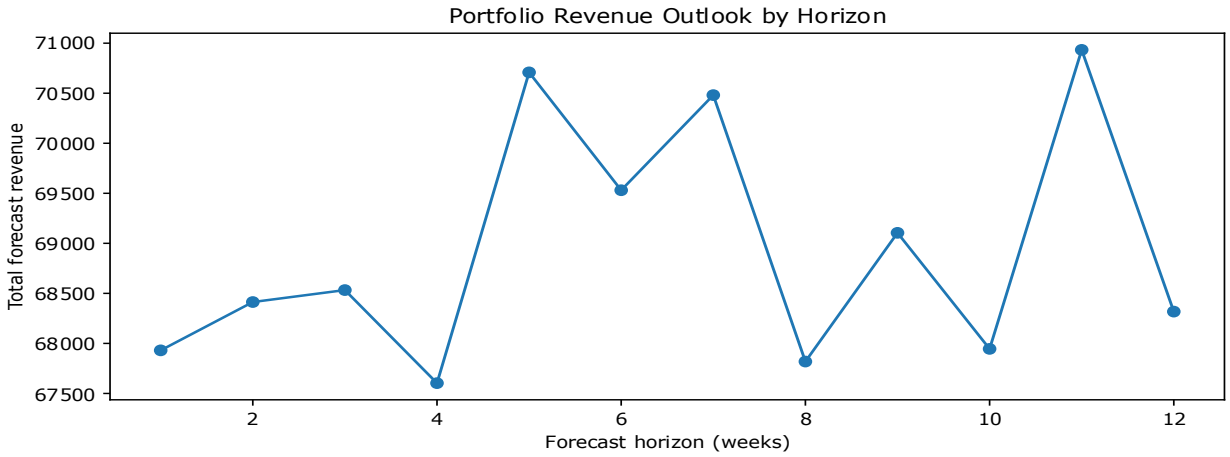


Figure A1: Total projected portfolio revenue by forecast horizon. Weekly revenue remains within a relatively narrow range across the 12-week projection window.

Appendix A2: Block-Level Test MAPE by Volatility Group

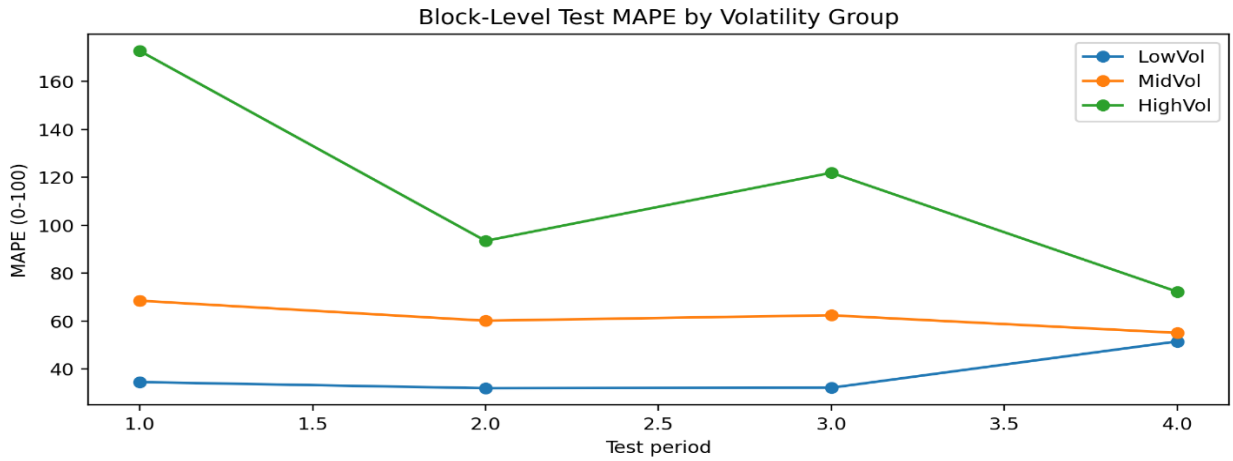


Figure A2: Held-out block-level test MAPE across the four equal-sized test periods, segmented by SKU volatility tercile. High-volatility SKUs exhibit the greatest error levels, while low-volatility SKUs remain the most stable